

ICE STORM & WINTER GRID FAILURE

What to do when the power goes out in a hard freeze — hypothermia and frostbite recognition and care, safe heat without electricity, and getting through the days that follow without becoming a second casualty of the storm.

Before you read further: This guide is general first aid and emergency-preparedness education. It is not a substitute for professional medical care, EMS response, or a certified first aid/CPR course, and it is not legal advice. In any true emergency, call **911** immediately. See the full disclaimer at the bottom of this book.

JUMP TO: [Recognize](#) → [Rewarm Safely](#) → [Heat Safely](#) → [Travel Decisions](#)

1. WHY THE FIRST HOUR (AND THE DAYS AFTER) MATTER

An ice storm that takes down the power grid is really two emergencies stacked on top of each other, and they move at completely different speeds. The first is fast: a body losing heat to cold, wet, or wind can develop dangerous hypothermia or frostbite within the first hour, sometimes faster, and the difference between mild cold discomfort and a medical emergency can come down to how quickly you recognize what's happening. The second is slow: the power may stay out for days, and the choices made in the first hour — how you heat your home, whether you drive anywhere, whether you check on the people nearby who need help — determine how safely you get through everything that follows.

This book treats both timelines as equally real. It isn't a standalone medical guide, and it isn't a standalone outage-preparedness guide — it's the intersection of the two, written specifically for the scenario where a hard freeze and a power outage hit at the same time, which is exactly when an ice storm tends to happen.

THE CORE IDEA

Cold injury is a fast clock; a winter outage is a slow one. Run the fast clock's response — get warm, get dry, recognize hypothermia and frostbite early — without ever losing sight of the slow clock's biggest single danger: generating heat indoors in a way that fills the air with carbon monoxide. Section 5 covers that danger in full, and it is not optional reading.

2. HYPOTHERMIA RECOGNITION

Hypothermia is what happens when the body loses heat faster than it can produce it, and core body temperature drops below the level needed for normal organ function. It doesn't require an arctic blast — a cold, wet, windy night in the 40s can cause it, especially indoors without heat, in wet clothing, or in an older adult or young child whose body regulates temperature less efficiently. Recognizing which stage someone is in changes how urgently you need to act.

STAGE	WHAT YOU'LL SEE	WHAT IT MEANS
Mild	Persistent shivering, cold and pale skin, mild confusion or clumsiness, faster breathing and heart rate	The body is still actively fighting to generate heat — this is recoverable with the steps in Section 3
Moderate	Shivering becomes intense, then starts to fade; slurred speech, stumbling, drowsiness, confusion or irritability, fumbling hands	Core temperature is dropping further; the body's ability to compensate is starting to fail
Severe	Shivering stops entirely, muscles become stiff or rigid, breathing and pulse become slow and weak or hard to find, the person becomes very confused or loses consciousness, skin may look blue or gray	A medical emergency — the body has stopped trying to warm itself, and cardiac arrest is a real risk

SHIVERING THAT STOPS IS NOT A GOOD SIGN.

It's tempting to read a person "calming down" after violent shivering as improvement. In moderate-to-severe hypothermia, it's the opposite: the body has exhausted its ability to generate heat through muscle activity and shivering shuts off. Treat a sudden stop in shivering — especially paired with confusion, slurred speech, or drowsiness — as a sign the person is getting worse, not better, and call 911 immediately if you haven't already.

Call 911 for any suspected moderate or severe hypothermia. For mild hypothermia in an otherwise healthy, alert adult, the steps in Section 3 combined with getting to a warm environment are usually sufficient — but if the person doesn't improve within a reasonable time, or you're unsure which stage you're looking at, treat it as the more serious case and call for help.

3. HYPOTHERMIA REWARMING TECHNIQUE

The instinct with someone who's dangerously cold is to warm them up as fast as possible — a hot shower, a heating pad pressed against the skin, a seat right in front of a space heater. For mild hypothermia in an alert, otherwise healthy person, gentle warming is fine. For moderate-to-severe hypothermia, rapid or aggressive rewarming is actively dangerous, not just unhelpful, and this section exists to make that distinction impossible to miss.

1 Get the person out of the cold and out of any wet clothing.

Move them somewhere sheltered from wind and precipitation. Cut clothing off rather than pulling it over the head/limbs if that causes rough movement — see the handling note below.

2 Dry the skin gently and wrap in dry insulation.

Dry blankets, sleeping bags, or dry clothing layered over the person, including under them if the ground or floor is cold — insulate from both directions.

3 Rewarm gradually, using the body's own heat and passive insulation

— dry blankets, body-to-body contact from another person (skin-to-skin under shared blankets, for severe cases, is a recognized field technique), a warm (not hot) room. The goal is a slow, even rise in core temperature, not a fast one.

4 If the person is fully alert, able to swallow normally, and shows no signs of moderate or severe hypothermia,

warm (not hot) sweet, non-alcoholic, non-caffeinated drinks can help. Do not give anything by mouth to anyone who is confused, drowsy, or has any trouble swallowing — the aspiration risk is real and the calorie benefit is not worth it.

5 Handle the person gently, and minimize rough movement or jostling,

especially in moderate-to-severe cases. A cold heart can be prone to a dangerous rhythm disturbance, and rough handling has been documented as a trigger. Move them as you would someone with a suspected spinal injury: careful, deliberate, no sudden repositioning.

6 Call 911 for any moderate or severe hypothermia

and continue passive rewarming while you wait. If the person has no detectable breathing or pulse after checking carefully for up to a minute — a hypothermic pulse can be extremely slow and faint — and you're trained in CPR, begin it and follow dispatcher instructions.

NEVER RAPIDLY OR AGGRESSIVELY REWARM MODERATE-TO-SEVERE HYPOTHERMIA.

No hot baths or hot showers. No heating pads, electric blankets on high, or any direct high-heat source placed against the skin. No vigorous rubbing or massaging of cold limbs. Rapid rewarming of someone with moderate-to-severe hypothermia can push cold, poorly-oxygenated blood from the extremities back to the heart all at once (sometimes called "afterdrop"), which can trigger a dangerous heart rhythm, and can burn skin that's too numb to feel the heat building up. Gradual, passive rewarming is the correct technique — every time, no exceptions for how cold or how anxious you feel to fix it fast.



1STHOUR

SAFE INDOOR HEATING

— NEVER RUN IT INDOORS. EVER. —

Carbon monoxide from a generator or grill is the leading cause of death in the aftermath of a winter power outage.



4. FROSTBITE RECOGNITION & FIELD CARE

Frostbite is the freezing of skin and underlying tissue, most common on fingers, toes, ears, nose, and cheeks — the parts of the body with the least insulation and blood flow. It progresses through recognizable stages, and how far it's progressed changes exactly what to do.



1ST HOUR

FROSTBITE STAGES

— RECOGNITION & FIELD CARE —

Never rub the area. Never use direct high heat. Rewarm gradually in warm water.

FROSTNIP

MILD
REDNESS



NO BLISTERING
Skin feels cold and numb

SUPERFICIAL FROSTBITE

PALE/WAXY
SKIN

MILD
SWELLING

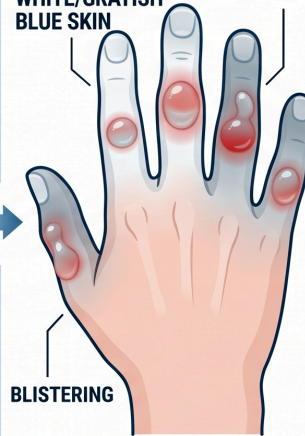


MILD SWELLING
Freezing of skin surface
& underlying tissue

DEEP FROSTBITE

HARD,
WHITE/GRAYISH-
BLUE SKIN

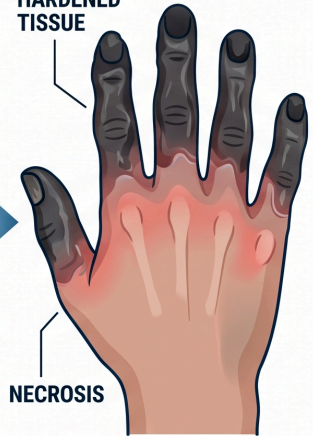
BLISTERING



BLISTERING
Damage to skin, nerves,
and muscle

SEVERE/NECROSIS

BLACKENED,
HARDENED
TISSUE



NECROSIS
Irreversible tissue death;
risk of amputation

1

FROSTNIP

Numbness, tingling, and redness. No ice crystals have formed in the tissue yet. Fully reversible with rewarming.

2

SUPERFICIAL

Skin looks pale, waxy, or grayish-white. May feel firm on the surface but soft underneath. Blistering with clear fluid can appear hours after rewarming.

3

DEEP

Skin is hard, cold, and feels wooden all the way through — not just on the surface. Blood-filled blisters can form. Significant tissue damage is likely.

4

SEVERE / NECROSIS

Tissue turns blackened and hardened as it dies. This stage needs emergency medical/surgical care — field rewarming alone will not reverse it.

WHAT TO DO

- 1 Get out of the cold and remove wet or constrictive clothing and jewelry**
from the affected area before it swells further.
- 2 Rewarm gradually in warm — not hot — water,**
roughly 99–102°F / 37–39°C (comfortably warm on your own unaffected skin, never hot to the touch). Use a thermometer if you have one; if not, err toward the cooler end of that range rather than guessing warmer. Soak the affected area continuously for 15–30 minutes until it feels warm and looks flushed/red, not until it's just "not cold" anymore.
- 3 Protect blisters and don't break them.**
Loosely cover the area with clean, dry, non-adherent dressing or cloth once rewarmed.
- 4 Get emergency medical care for any deep or severe frostbite,**
and for any superficial frostbite that doesn't clearly improve after rewarming. Frostbite that has caused blistering or hardened tissue needs a clinical evaluation — this is not a wait-and-see injury past the frostnip stage.

NEVER DO THESE — THEY CAUSE REAL, ADDITIONAL TISSUE DAMAGE:

Never rub or massage a frostbitten area, with your hands, snow, or anything else — ice crystals have formed inside the tissue, and rubbing grinds them against cells, causing mechanical damage on top of the cold injury. Never use direct high heat — a stove, campfire, car heater vent, or heating pad pressed against the skin — frostbitten tissue is numb and won't signal pain until it's already burned. Never rewarm the area if there's a realistic chance it will refreeze before you can get the person to definitive care — thawing and then refreezing causes dramatically worse tissue damage than staying frozen a while longer. If refreezing is a real risk (you still have to travel through the cold to reach help), keep the area protected and frozen, and get to care rather than rewarming in the field.

Frostbite and hypothermia often occur together — check for both whenever you find one.

5. SAFE INDOOR HEATING WITHOUT POWER

THIS IS THE SINGLE MOST IMPORTANT SECTION IN THIS BOOK — THE SAME STANDARD AS GRID-DOWN'S OWN GENERATOR-SAFETY SECTION, AND IT DOES NOT SOFTEN FOR WINTER.

Carbon monoxide (CO) from an improperly placed generator, grill, or camp stove is the leading cause of death in the aftermath of winter power outages specifically, because losing heat in a hard freeze creates enormous pressure to find warmth fast, indoors, by any means available. CO is colorless, odorless, and tasteless. You cannot detect it forming. The only reliable protection is never creating the conditions for it to build up around you in the first place.

NEVER RUN THESE INDOORS, FOR ANY REASON, EVEN BRIEFLY:

- **A gasoline, propane, or diesel generator** — not in a basement, not in a room with the windows open, not "just for a minute" to warm up.
- **A charcoal or propane grill** — indoors, in a garage, or on an enclosed porch. Grills produce CO exactly like generators do.
- **A camp stove or portable propane heater rated for outdoor use only** — check the rating before assuming any fuel-burning device is safe indoors; most are not.
- **A gas stove or oven used to heat a room.** This feels like an obviously available option during a cold outage, which is exactly why it's a leading cause of indoor CO poisoning during them.
- **Any of the above in a garage or shed, even with the door fully open.** An open garage door does not ventilate exhaust fast enough — CO can accumulate to lethal levels inside the garage and drift into the attached house.

✓ **SAFE INDOOR OPTIONS**

- Layered clothing, blankets, and sleeping bags — the first line of defense, before any heat source
- Closing off and gathering in one small, well-sealed room rather than heating the whole house
- Battery-powered or rechargeable electric space heaters and heated blankets, used per their manufacturer instructions, away from flammable material
- A permanent, professionally installed, code-compliant fireplace or wood stove that was already part of the home before the outage — never one improvised during one — with a clear, unblocked flue and a working CO detector nearby

✗ **NEVER INDOORS**

- Generators, grills, and camp stoves — see the list above
- Gas ovens/stovetops used as room heaters
- Any device rated "outdoor use only"
- An improvised fire, fire pit, or chiminea brought inside — even temporarily

SAFE PLACEMENT FOR ANY FUEL-BURNING DEVICE YOU DO USE

- 1 Place it outdoors, in the open air, at least 20 feet from your home**
in every direction, and at least 20 feet from any neighboring structure — further if wind is blowing exhaust toward a building.
- 2 Point exhaust away from your home and any windows, doors, or vents**
— yours or a neighbor's.
- 3 Never run it under an awning, tarp, or any covered structure**
even if it's freestanding and outdoors.
- 4 Install battery-powered or battery-backup carbon monoxide detectors inside your home**
— on every level, near sleeping areas — and test them before you need them.

- 5** If a CO alarm sounds, get everyone outside into fresh air immediately and call 911 — do not go back inside to search for the source or open windows first.

IF YOU SMELL EXHAUST OR FEEL CO-POISONING SYMPTOMS

Headache, dizziness, weakness, nausea, confusion, or chest pain in anyone in or near a heated space is a medical emergency. Get everyone into fresh air immediately, call 911, and do not re-enter until the space has been checked and cleared.

This guidance is the same standard used in the [Grid-Down / Extended Power Outage](#) protocol's own generator-safety section — the rule doesn't change because the outage happens to be caused by ice instead of a storm or grid failure from another cause. If you've read that book, treat this as the winter-specific extension of the exact same rule, not a different one.

6. DRIVING & TRAVEL DECISIONS IN ICE CONDITIONS

Most winter-storm injuries and deaths that aren't cold-injury-related happen in vehicles. The single best decision is often the one that doesn't involve driving at all.

WHEN TO STAY OFF THE ROADS ENTIRELY

- Any active ice storm warning or advisory covering your route
- Visible ice on trees, power lines, or your own vehicle's exterior surfaces before you've even started the engine
- Any trip that isn't genuinely necessary — "I can probably make it" is not the same question as "do I need to go"
- Bridges and overpasses specifically — they freeze first and lose traction before the rest of the road surface does, even when the road itself looks clear

IF TRAVEL IS UNAVOIDABLE, KEEP THIS IN THE VEHICLE

☐ **Blankets and extra warm layers**
— for everyone in the vehicle, in case you're stranded.

☐ **Food and water**
— enough to last well beyond your expected travel time.

☐ **Phone charger or battery power bank**
— a dead phone removes your ability to call for help.

☐ **Flashlight and extra batteries**
— for visibility and signaling.

☐ **Sand, salt, or cat litter, plus a small shovel**
— for traction if you get stuck.

☐ **Ice scraper and jumper cables**
— basic winter-driving equipment that's easy to forget until you need it.

IF YOU BECOME STRANDED

1 Stay with the vehicle.

It's far easier for rescuers to spot a car than a person on foot in a storm, and the vehicle provides shelter from wind and precipitation.

2 Call or text 911 with your location

as precisely as you can describe it, and let them know you're staying with the vehicle.

3 If you run the engine for heat, first make sure the exhaust pipe is completely clear of snow and ice.

A blocked tailpipe can force carbon monoxide back into the cabin — the same CO risk from Section 5, just in a smaller enclosed space. Check the exhaust pipe before every time you restart the engine, not just once.

4 Run the engine and heater for about 10 minutes each hour

to conserve fuel and limit exhaust exposure, and crack a window slightly whenever it's running for additional ventilation.

5 Signal your location.

Tie a brightly colored cloth to the antenna or a partially rolled-down window, and use hazard lights sparingly to conserve the battery.

6 Move your arms and legs periodically to keep circulation going

without leaving the vehicle, and huddle together with other passengers for shared body heat.

7. CHECKING ON VULNERABLE NEIGHBORS

A winter grid-down event puts some people at dramatically higher risk than others, and they're often within a few doors of you. Knowing who's vulnerable before the outage happens — and having a plan to check on them — is one of the highest-leverage things you can do in the first hour and the days after.

- **Older adults living alone** often have less internal heat-regulation capacity, may not notice or report how cold their home has gotten, and may be reluctant to ask for help. A knock on the door costs you nothing.
- **Anyone medically dependent on powered equipment** — oxygen concentrators, home dialysis, CPAP machines, powered wheelchairs, refrigerated medication — faces a second emergency layered on top of the cold one the moment their backup battery runs out. If you know a neighbor fits this description, find out their plan (backup battery runtime, whether they need to relocate to a powered facility) before the outage, not during it.
- **Families with infants or young children** — see Section 8 for why infants are at elevated hypothermia risk specifically.
- **Anyone without a reliable way to call for help** if their phone dies and they have no backup power — a simple in-person check may be the only way anyone finds out they need it.

HOW TO CHECK SAFELY

A knock on the door and a few minutes talking through it or on the phone is often enough to confirm someone is warm, has supplies, and has a way to reach help if things get worse. If a vulnerable neighbor isn't answering and you have reason to believe they're home and possibly in danger, call 911 or your local non-emergency line for a welfare check rather than forcing entry yourself.

8. SPECIAL SITUATIONS

INFANTS

Infants lose body heat much faster than adults relative to their size, and can't generate heat by shivering the way older children and adults do. They also can't tell you they're cold. Watch for cold-to-the-touch skin (especially the chest and abdomen, not just the hands and feet, which are cold on a healthy infant too), unusual lethargy, weak cry, or refusal to feed — any of these in a cold environment warrants immediate warming and, for anything beyond mild chill, a call to 911 or the infant's pediatrician.

OLDER ADULTS

In addition to the reduced heat-regulation capacity noted in Section 7, some medications commonly used by older adults can mask the early signs of hypothermia (reduced shivering response) or affect judgment about how cold they actually are. Don't rely on an older adult's own self-report of "I'm fine" as the only signal — check skin temperature and mental clarity directly.

PETS

Bring pets indoors during an ice storm outage whenever possible — the same cold that endangers people endangers them, and "they have fur" is not reliable protection for most household pets in genuinely cold conditions. If a pet must go outside briefly, limit exposure time, and check paws for ice buildup or road-salt irritation afterward. Never leave a pet in a parked vehicle in cold weather — a car loses heat quickly and offers no more protection than the outdoors once the engine is off.

9. ESSENTIAL ITEMS CHECKLIST

Everything in this protocol assumes certain supplies are within reach before the cold and the outage hit at the same time. Run through this list before you need it.

☐ **Extra blankets and sleeping bags rated for cold weather**

— Section 3. The core tool for passive, gradual rewarming.

☐ **Battery-powered or rechargeable space heater**

— Section 5. The only heat source with zero CO risk.

☐ **Battery-powered carbon monoxide detector(s)**

— Section 5. Non-negotiable if you own or might ever use any fuel-burning device.

☐ **Warm water source or way to heat water safely**

— Section 4. For frostbite rewarming at a body-safe temperature.

☐ **Battery bank / phone charger**

— Sections 6 & 7. Keeps you able to call 911 or check in on neighbors.

☐ **Vehicle winter kit (blankets, food, water, sand/litter, ice scraper)**

— Section 6. Assemble it before the first storm of the season, not during one.

☐ **Flashlight and extra batteries**

— Section 6. For visibility, signaling, and checking on neighbors after dark.

☐ **Layered cold-weather clothing for every household member, including infants**

— Sections 3 & 8. The first line of defense before any heat source.

WHERE THESE LIVE

Keep cold-weather gear and your CO detector together and checked before the first freeze of the season — under real cold-injury pressure, you don't want to be hunting for a blanket in a dark, powerless house.

10. REGIONAL & CLIMATE NOTES

Ice storms and the winter grid failures that follow them are overwhelmingly a concern for northern and upper-Midwest states and any region where freezing rain, not just snow, is a regular winter feature — the freezing rain is what coats power lines and tree limbs in ice heavy enough to snap them, which is the actual mechanism behind most ice-storm outages. This book doesn't attempt a full 12-state Good Samaritan grid the way the pilot's does — geography here matters as climate risk, not legal variation, so it's framed by region instead.

Primary risk zone: the northern tier and upper Midwest (states such as Minnesota, Wisconsin, Michigan, upstate New York, New England, and the northern Great Plains) see ice storms and extended winter outages most frequently, with infrastructure generally built to expect them.

Secondary risk zone — arguably higher per-event danger: the mid-South and mid-Atlantic (Texas, Oklahoma, the Carolinas, Virginia, Tennessee) see ice storms less often, but with less winterized infrastructure and less public familiarity with cold-injury risk, a single severe event can cause outsized harm — this is exactly the region that produced some of the deadliest recent ice-storm grid failures.

Wherever you are, the medical content in Sections 2–4 and the CO-safety rule in Section 5 don't change by region — only how likely you are to need them, and how prepared your local infrastructure is to help you, does.

11. RELATED PROTOCOLS

A winter grid-down event often overlaps with other first-hour emergencies. These protocols pick up where this one leaves off.



Grid-Down / Extended Power Outage

The full operational playbook this book's Sections 1 and 5–7 extend with a winter-specific lens — food safety, water, and communication when the power's been out for days.

LIVE



Shock Management

Severe hypothermia is itself a shock presentation — if positioning, warmth, and monitoring in Section 3 aren't enough, this is the deeper protocol.

LIVE

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12. MEDICAL DISCLAIMER

This guide is for general educational purposes only and does not constitute medical advice, diagnosis, or treatment, emergency management guidance, or legal advice. It is not a substitute for professional medical care, emergency medical services, your local emergency management agency's official guidance, or a certified first aid, CPR, or emergency-preparedness course, which we strongly recommend taking in person.

In any real emergency, call 911 (or your local emergency number) immediately. Nothing in this guide should delay that call.

Generator, carbon monoxide, and indoor-heating safety guidance in this book reflects general best practices (consistent with CDC and Ready.gov consumer guidance) but does not replace your specific equipment's manufacturer instructions, a licensed electrician's or HVAC professional's guidance for any permanent installation, or your local fire code.

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Editor's note: this edition reflects content as of August 25, 2026. 1st Hour periodically revises protocol guides as best practices evolve — visit 1sthourprotocols.com/iceover-event for the current edition and to re-download the latest PDF.

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